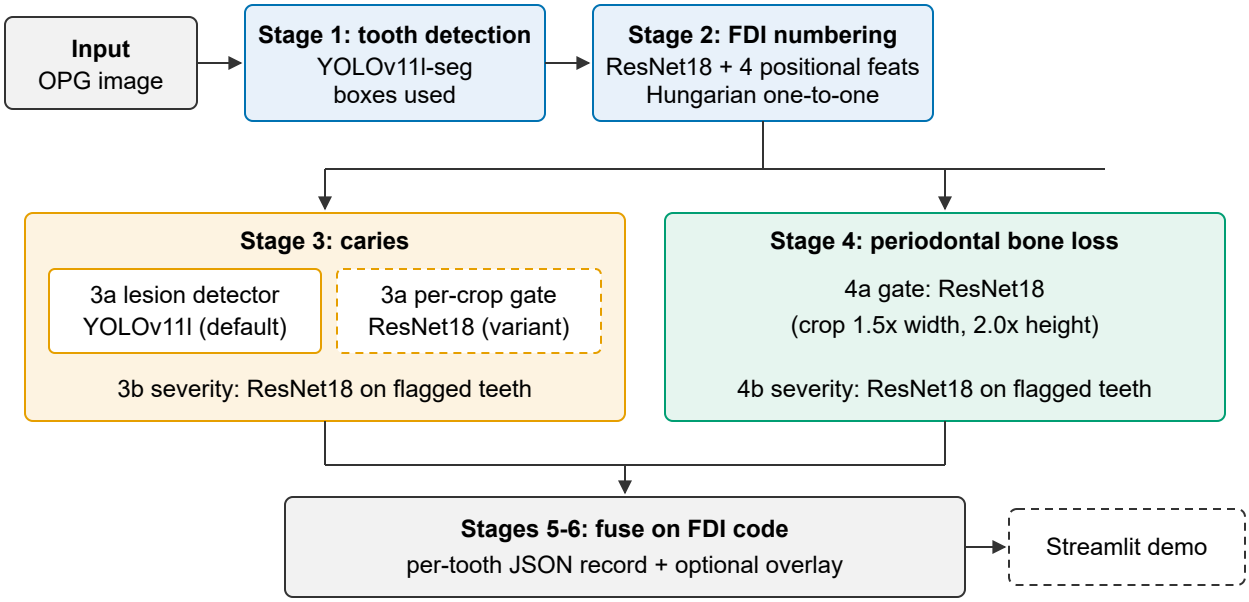
Input
OPG image
Stage 1: tooth detection
YOLOv11l-seg
boxes used
Stage 2: FDI numbering
ResNet18 + 4 positional feats
Hungarian one-to-one
Stage 3: caries
3a lesion detector
YOLOv11l (default)
3a per-crop gate
ResNet18 (variant)
3b severity: ResNet18 on flagged teeth
Stage 4: periodontal bone loss
4a gate: ResNet18
(crop 1.5x width, 2.0x height)
4b severity: ResNet18 on flagged teeth
Stages 5-6: fuse on FDI code
per-tooth JSON record + optional overlay
Streamlit demo